QING Bowen *(Weber)*

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Singapore PR, LLM & CV Senior Engineer (2 years exp.)

T-shaped with deep expertise in LLM optimizing and integration; broad proficiency in AI Application and team management.

Education

Nanyang Technological University (NTU) — Aug 2022 – Dec 2023, *Singapore*
***M.Sc. (Artificial Intelligence)**, Advisor: Prof. Dusit Niyato, Concentration: CV & AIGC*

Southern University of Science and Technology (SUSTech) — Sep 2018 -Jun 2022, *Shen Zhen, China*
***B.Eng. (Computer Science)**, Concentration: Intelligent Perception, Outstanding graduates*

National University of Singapore (NUS) — Aug 2021 - Dec 2021 & Jul 2019, *Singapore*
Exchange Student** (Research with Full Scholarship) **CAP 5.0/5.0

Research Experience

Master's Project: AIGC-based Enhancement of Anomalous Data for Autonomous Driving — Jun 2023 - Oct 2023

- **Employed Stable Diffusion** with **LoRA** fine-tuning, integrated **Segment Anything** ground segmentation and **ControlNet** Canny contour preservation, leveraging CARLA Simulator data for reflection enhancement, boosting autonomous parking system safety.

Paper submission: Structured Runtime Safety Monitoring for Camera-Based Driver Monitoring Systems — IEEE, ITSC 2026

- Developed a hybrid **YOLOv11-Qwen3VL** L1–L2 safety monitor hierarchical pipeline for cabin DMS, using CoT-based reasoning as cross-verification, achieving **92.16%** accuracy in **33ms-real-time** on the YawDD Driver sleep apnea detection dataset.

Work Experience

Desay SV Automotive — Apr 2024 - On-going, *Jurong East, Singapore*
LLM & CV Senior Engineer

1 Vertical AI research: Autonomous driving/In-cabin multi-modal LLM development Mar 2025 - On-going

- Mining driving training video using local-hosted **Qwen2.5-VL:32B**, **Amap API**, and **YOLOv8** to filter long-tail-scenes and odd-objects.
- Use **OpenAI/Gemini API** to automatically annotate traveling recording clip with caption, then **full-parameter SFT** the Qwen2.5-VL:3B.
- Dataset prepare and VLM fine-tune for **designated driver module**, including clothing, unusual behavior, and lost item detection.
- Use Qualcomm QNN toolchain to **quantize** Qwen3-VL:2B into W8A16 and **operator adaptation** on 8797/8295 embedded platform.
- Developed vehicle-side **Joint-BERT based NLU** module to classify passenger's intent and **extract built-in-tool-call slot.**

2 Horizontal AI application: Internal Efficiency Boost AI Agent Development Sept 2024 - On-going

- **Advise multiple (10+) interns** and deployed enterprise-grade LLM toolchain with **Qwen3:30B-A3B** and **Deepseek-R1:671B** via **vLLM**.
- Deliver **10+ internal productivity agents**, e.g. News/Feishu task Agent, Doc compare, Invoice pipeline. Serving R&D, HR, UI/UX, F&A Dept.
- Using an **OpenClaw with Gemini** to autonomous closed-loop perception and control robotic arm to insert plugs in ISSIC Lab simulator.
- Applied Feishu Aily workflow in **HR talent pool** for resume scoring and win Hackathon prize.
- Responsible as an **administrator** for company's A100 GPU server maintenance and H200 GPU server procurement and selection.

3 Legacy CV perception project: BEV Map traffic-lane auto-labeling model Apr 2024 - Mar 2025

- Developed BEV lane & marker detection models based on **GeMap** and **VMA** with custom loss functions, e.g. **Point2Line Loss**
- Designed streaming **2D crop augmentation** for 4D high-res (up to **16k×8k**) maps, reducing **GPU memory usage** by 40%.
- Enabling stable single batch on 8-GPU **pipeline-parallel** training with **Deepspeed** distribute framework.
- Boosting **chamfer-based mAP** and improving curve smoothness in BEV outputs. Improving detection accuracy by 7% mAP.
- Built **RGB** + **LiDAR intensity fusion net**; explored pre/mid/post fusion via **Transformer**, achieving +5% recall on rare traffic signs.

4 University-Enterprise-Collaborate project: Diffusion-based driving World model Jan 2025 - Mar 2025

- School-enterprise collaboration with **NTU**, **SYSU**, generating **ego multi-camera videos** based on signals from a **global traffic manager**.

Innovus Capital — Aug 2023 - Jan 2026, *One-north, Singapore*
Startup Co-founder & CTO

1 Founded a financial market investment advisory firm and legally holding share in Singapore

- Deployed & maintained company website (**innovus.ltd**) on public cloud infrastructure with automated monitoring and backups.
- Deployed **Qwen3:7B** LLM locally to perform financial analysis, eliminating reliance on external API calls.
- Integrated market data APIs (**Finhub**, **YFinance**) to fetch real-time U.S. stock data for internal analytics.
- **Recruited & managed** full-time employees to support both business operations and technical development.
- Built automated LLM-powered reporting pipeline incorporating: sentiment analysis of financial news/media, quantitative modeling using **structured market analysis**, and structured generation of comprehensive market intelligence reports.

Internship Experience

Desay SV Automotive (Return offer with signing bonus) — Jun 2023 - Oct 2023, *Jurong East, Singapore*
AIGC Algorithm Intern

Tencent — Mar 2021 - Aug 2021, *Shen Zhen, China*
CV Algorithm (VIQUAL Ophthalmology Public Welfare Project)

DJI Innovation — Feb 2022 –Jun 2022, *Shen Zhen, China*
SLAM Test & Development (School-Enterprise Cooperation)

Miscellaneous

Python, Java, C# *Proficient* **C++, SQL, HTML** *Intermediate* **English, Mandarin** *Business* **Japanese** *Elementary*

Skill: **LLM Agent, RAG workflow, Prompt Eng. MCP**, VLLM, Dify, RAGflow, Coze, Docker, Git, Linux, Adobe (Ps, Pr, Au)

Interests: **AIGC(SORA, Diffusion)**, MMD Animation, VR/AR/XR, Multimedia Processing, Graphic Design, Photography